Plan

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Two lectures on the OUU paper:

- Value-based uncertainty and multi-objective optimization
- Functorial uncertainty

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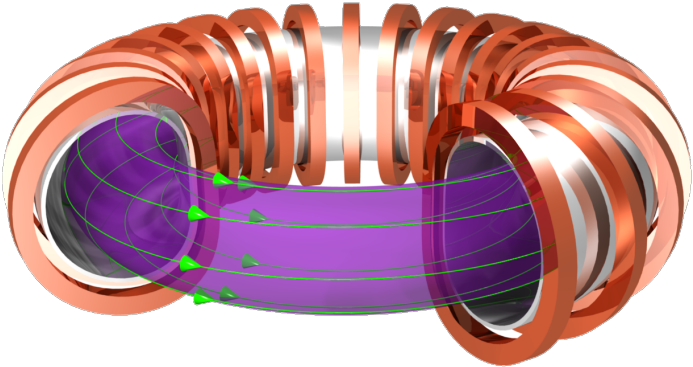
- Functorial uncertainty

Today:

- Value-based uncertainty and multi-objective optimization

A bit of motivation . . .

Motivation: plasma physics, tokamaks



Motivation: DREAM simulator

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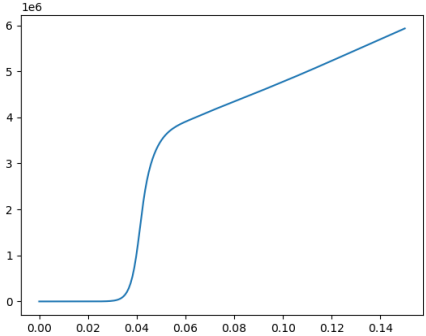
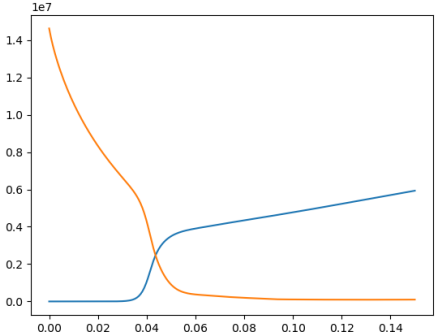
- DREAM (*Disruption Runaway Electron Analysis Model*) simulates the outcome of a disruption experiment (including, among others, the runaway electron distribution function, the current quench, and the thermal quench) self-consistently...
- In simple terms, DREAM is a function that maps the input and control parameters (which fully specify a disruption experiment at a 1D, 2D, or 3D layer) to a possible outcome of that experiment.

Motivation: bad outcome

- Runaway current (iRE): starts at 0, grows exp. to 4 MA within 0.05 s, then grows linearly.
- Ohmic current (iOhm): starts at 15 MA, decays to 1% within 0.05 s.



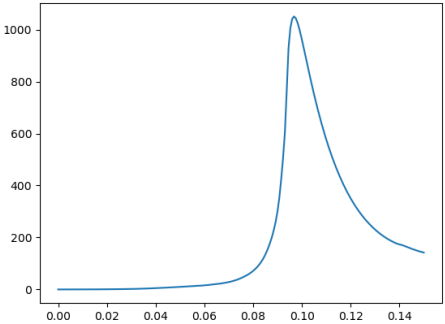
runaway current (max 6 MA)



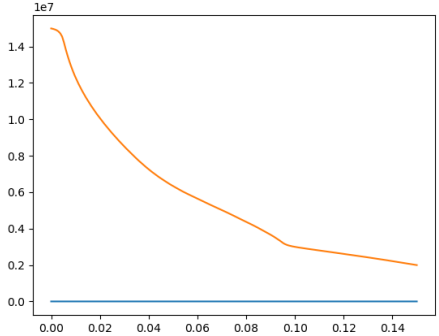
ohmic current (max 15 MA)

Motivation: good outcome

- Runaway current (iRE): starts at 0, grows exp. to 1 kA within 0.05 s, then decreases exponentially.
- Ohmic current (iOhm): starts at 15 MA, decays to 7 MA within 0.05 s.

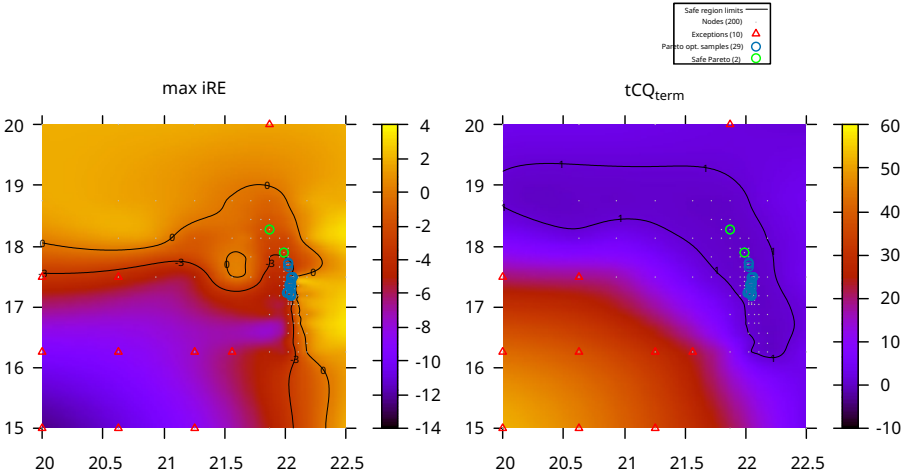


runaway current (max 1 kA)



ohmic current (max 15 MA)

Motivation: low iRE ⇒ high thermal quench (tCQ)!



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- X is a subset of $\mathbb{R}^n$ in which the system can be operated, and X_σ is the subset of desirable or safe outcomes.

Multi-objective optimization in physics, engineering, climate science

- Ideally, one would like to find controls $ys \subseteq Y_\varphi$ that minimize all $f_1 \ldots f_n$.

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Exercise 1.1

Specify what it means for a control $y \in Y_\varphi$ to be safe using X_σ.

Exercise 1.2

Specify what it means for $y \in Y_\varphi$ to yield results $f_1\ y \ldots f_n\ y$ such that one cannot improve a result without making some other result worse (in the context of minimization).

Single objective minimization (finite case)

- Finding all the elements of a finite set at which a function attains its minimum:

$$
\begin{aligned}
&\mathit{minOf} &&:: \mathit{Ord}\ b \Rightarrow (a \to b) \to \mathit{Set}\ a \to b \\
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  1. $\forall\ a \in as.\ \ \mathit{minOf}\ f\ as \leqslant f\ a$
  2. $\forall\ a \in as.\ \ a \in \mathit{argminOf}\ f\ as \Rightarrow f\ a = \mathit{minOf}\ f\ as$

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3. $\forall\ a \in as.\ \ f\ a = \mathit{minOf}\ f\ as \Rightarrow a \in \mathit{argminOf}\ f\ as$

Single objective minimization (finite case)

Exercise 1.3

Make sure that you fully understand the specifications for *minOf* and *argminOf*. What is the type of *map f as* for a function $f : \mathbb{R}^m \to \mathbb{R}$? What is the type of *argminOf f as*? Can we apply *minOf* to functions of type $\mathbb{R}^m \to \mathbb{R}^n$?

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Exercise 1.4

The implementation

$$
\begin{aligned}
\mathit{minOf}\ f\ \mathit{as} &= \mathit{minimum}\ (\mathit{map}\ f\ \mathit{as}) \\
\mathit{argminOf}\ f\ \mathit{as} &= \textbf{let}\ m = \mathit{minOf}\ f\ \mathit{as}\ \textbf{in}\ \mathit{filter}\ (\lambda\ a \to f\ a == m)\ \mathit{as}
\end{aligned}
$$

can be tested to fulfill the specifications of *minOf* and *argminOf* (see *test11*, *test12*, and *test13* in the paper). What is the type of *minimum*? What is the type of *filter*? Implement and run *test11* for a function of your choice of suitable type.

Multi objective minimization: Pareto optimality

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Exercise 1.5

What is the type of $\prec$? Specify what it means for a subset of X to be indifferent with respect to $\prec$. Define $\prec$ for $X = \mathbb{R}^2$ consistently with Fig. 1 of the paper. Define the relation *indiff* $(\prec)$, the "indifference relation" computed from $(\prec)$. Prove that *indiff* $(\prec)$ is not transitive.

Multi objective minimization (finite case)

- As in SOM, one can easily compute the Pareto optimal subset (Pareto front) of a finite set for a given dominance relation $\prec$:

$$\mathit{paretoOpt} :: (\mathit{NDim}\ n, \mathit{Ord}\ b) \Rightarrow \mathit{Set}\ (n\ b) \rightarrow \mathit{Set}\ (n\ b)$$

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- The idea is to define *paretoOpt* inductively:

$$\begin{aligned} \mathit{paretoOpt}\ [] &= [] \\ \mathit{paretoOpt}\ (x : \mathit{xs}) &= \mathit{bump}\ x\ (\mathit{paretoOpt}\ \mathit{xs}) \end{aligned}$$

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Exercise 1.7

If *bump* fulfills three specific requirements, then *paretoOpt xs* will correctly compute the Pareto optimal subset of *xs* for an arbitrary *xs*. Formulate these three requirements.

Multi objective minimization (finite case)

Exercise 1.8

Define $bump\ x$ for the empty set ($Set\ =\ List$).

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Exercise 1.9

Define $bump\ x$ for a set (list) of Pareto optimal points $p\ :\ ps$.

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Define $bump\ x$ for a set (list) of Pareto optimal points $p\ :\ ps$.

Exercise 1.10

Define a relation $isParetoOptOf :: (NDim\ n, Ord\ b) \Rightarrow Set\ (n\ b) \rightarrow Set\ (n\ b) \rightarrow Bool$ and test that, with your definition of $bump$, $paretoOpt\ xs$ indeed yields the Pareto optimal subset of xs for an arbitrary xs.

Multi objective minimization (finite case)

Exercise 1.11

Implement

$$\begin{aligned} &\mathit{paretoMinOf} &&:: (\mathit{NDim}\ n, \mathit{Eq}\ a, \mathit{Ord}\ b) \Rightarrow (a \to n\ b) \to \mathit{Set}\ a \to \mathit{Set}\ (n\ b) \\ &\mathit{argparetoMinOf} &&:: (\mathit{NDim}\ n, \mathit{Eq}\ a, \mathit{Ord}\ b) \Rightarrow (a \to n\ b) \to \mathit{Set}\ a \to \mathit{Set}\ a \end{aligned}$$

using *paretoOpt* alongside the *map*, *filter*, and membership operations available on $\mathit{Set} = \mathit{List}$.

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Exercise 1.12

Use *isParetoOptOf* from exercise 1.10 and $\mathit{Set} = \mathit{List}$ membership to write formal specifications for *paretoMinOf* and *argparetoMinOf* that are analogous to the specifications for *minOf* and *argminOf*.